Affine character sheaves via topological field theory and the orbit method

David Nadler, Zhiwei Yun

Lecture 1 (DN)

G conn. red $/\mathbb{C}$ (often for simplicity, assume G s.c.)

Today. State & discuss thm.

Def. Universal affine Hecke cat.

$$\mathcal{H} = \underset{\text{aka } \mathcal{N}}{Sh_{bimon}} (I^+ \backslash LG / I^+)$$

$$= \bigcup_{w \in \tilde{W}} Sh_{bimon} (I^+ \backslash LG^{\leq w} / I^+)$$

$$\cup$$

finite Hecke cat $\quad \mathcal{H}_f = \underset{\text{aka } \mathcal{N}}{Sh_{bimon}} (N \backslash G / N)$

Reminder. $\mathcal{C}$ sym mon. cat

$\cup$

A alg obj.

Def. Cocentre / Hochschild obj.

$$\begin{matrix} A \\ \downarrow tr \\ hh(A) = A \underset{A \otimes A^{op}}{\otimes} A \end{matrix}$$

Ex. $A = \mathcal{H}^f$

$$\frac{G}{G} \leftarrow \frac{G}{B} \rightarrow \frac{N \backslash G / N}{T}$$

$$\begin{matrix} \mathcal{H}^f & \xrightarrow{tr} & hh(\mathcal{H}^f) \\ \| & & \| \\ Sh_{\mathcal{N}}(N \backslash G / N) & \xrightarrow{ch} & Sh_{\mathcal{N}}\left(\frac{G}{G}\right) \end{matrix}$$

"derived char. sheaves"

Pseudo def "affine char shvs" $= hh(H)$ cocentre of AHC.

(so all depth 0 ...)

Rem 1) Cocentre $hh(A)$ only depends on $Mod_A(\mathcal{C})$

2) TFT interpretation.

1d $\quad \bullet \rightsquigarrow Mod_A(\mathcal{C})$

$\bigcirc \rightsquigarrow hh(A) = \int_{S^1} A$

Goal. geom. descrip of $hh(H)$ in terms of $Bun_G(E)$ ← genus 1

From local to global.

$$I^+ \backslash LG / I^+ = Bun_{G,N}\left(\left(D \coprod_{D^\times} D, \{0_-, 0_+\} \right) \right)$$

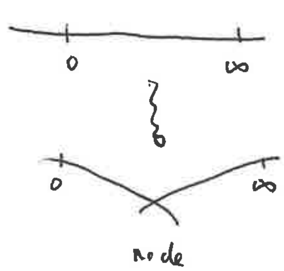
$$I_0^+ \backslash G[t,t^{-1}] / I_\infty^+ = Bun_{G,N}\left(\left(\mathbb{A}^1 \coprod_{\mathbb{G}_m} \mathbb{A}^1, \{0, \infty\} \right) \right)$$

Radon equiv. $\quad H = Sh_N(I^+ \backslash LG / I^+) \simeq Sh_N(I_0^+ \backslash G[t,t^{-1}] / I_\infty^+)$

convolution $\quad$ mon cats $\quad$ "bubbling" $\simeq H^{bub}$

Ψ^L left adj. to Ψ = nearby cycles. $\quad$ Bubbling Hecke cat.

(N. - Yun, Automorphic gluing)



$\mathbb{P}^1 \qquad Bun_{G,N}(\mathbb{P}^1, \{0,\infty\}) \qquad Sh_N(\cdots)$

Ψ = nearby cycles

$\mathbb{P}^1 \vee \mathbb{P}^1 \qquad Bun_{G,N}(\mathbb{P}^1 \vee \mathbb{P}^1, \{0,\infty\}) \qquad Sh_N(\cdots)$

Need enhancement $\quad$ Pass to $\mathbb{P}^1/\mu_k$

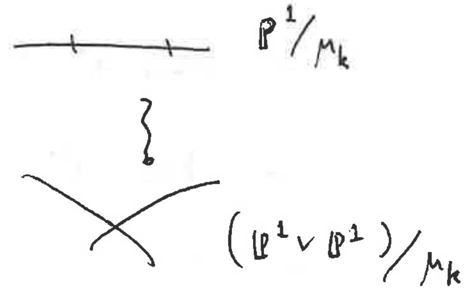
Given $\Sigma \in \mathrm{Bun}_G(\mathbb{P}^1/\mu_k)$

$\rightsquigarrow \Sigma|_0, \Sigma|_\infty \in \mathrm{Bun}_G(pt/\mu_k) \simeq \frac{1}{k}X_*(T)/\tilde{W}$

If we fix $\eta \in \frac{1}{k}X_*(T)/\tilde{W}$ generic, then $\mathrm{Bun}_G(\mathbb{P}^1/\mu_k)_\eta$ gives Iwahori level str. at stacky points

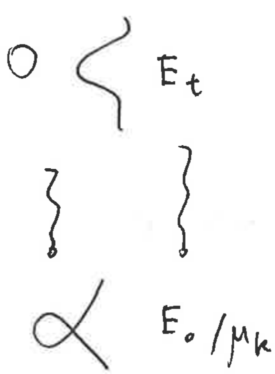
Rewrite $H^{bub} = Sh_N(\mathrm{Bun}_G(\mathbb{P}^1/\mu_k)_\eta)$

Now $\Psi: H^{bub} \longrightarrow H^{bub} \underset{Sh_N(T)}{\otimes} H^{bub}$



$\xleftarrow{\Psi^L}$ corresponds to prod of H

Consider "Tate curve":



$$\begin{array}{ccc} \mathrm{Bun}_G(E_t) & & Sh_N(\mathrm{Bun}_G(E_t)) \\ \rightsquigarrow & & \downarrow \Psi \\ \mathrm{Bun}_G(E_0/\mu_k)_\eta & & Sh_N(\mathrm{Bun}_G(E_0/\mu_k)_\eta) \\ & & \| \\ & & hh(H^b_T, H) \\ & & \| \\ & & Sh_N(T) \end{array}$$

Th (Li-N-Yun)

$$H \xrightarrow{tr} hh(H) = hh(H, H)$$

$$Sh_N(\mathrm{Bun}_G(\mathbb{P}^1/\mu_k)_\eta) = H^{bub} \;(= H) \to hh(H^b_T, H) \to hh(H,H) \simeq Sh_N(\mathrm{Bun}_G(E_t)) \simeq A_G(E_t)$$

$$H^{bub} \to hh(H^b_T, H) = Sh_N(\mathrm{Bun}_G(E_0/\mu_k)_\eta) \xrightarrow{\Psi^L} Sh_N(\mathrm{Bun}_G(E_t))$$

Key str.

$$\mathrm{Bun}_G(E_t) = \coprod_\lambda \mathrm{Bun}_P^{\lambda, ss}(E_t) \rightsquigarrow \text{semi-orthogonal decomp. of } \mathcal{A}_G(E_t).$$

Harder - Narasimhan

strat

Key str. 1) Construct analogous SOD on $hh(\mathcal{H})$

2) Check Ψ^L respects SOD

3) Prove Ψ^L is an equiv. on assoc. gr.

Lecture 2. Th (Li - Nadler - Y.)

$$hh(\mathcal{H}) \simeq Sh_{\mathcal{N}}(\mathrm{Bun}_G(E))$$

$$\mathcal{H} = Shv_{bimon}(I^+ \backslash LG / I^+)$$

Today. - $hh(\mathcal{H})$ has SOD indexed by $\nu \in \mathcal{N}(G)$: Newton points

- $hh(\mathcal{H})_\nu$ can be described using (twisted) CS on finite-dim'l gps.

Why? - $\mathrm{Bun}_G(E)^\nu$ HN strata

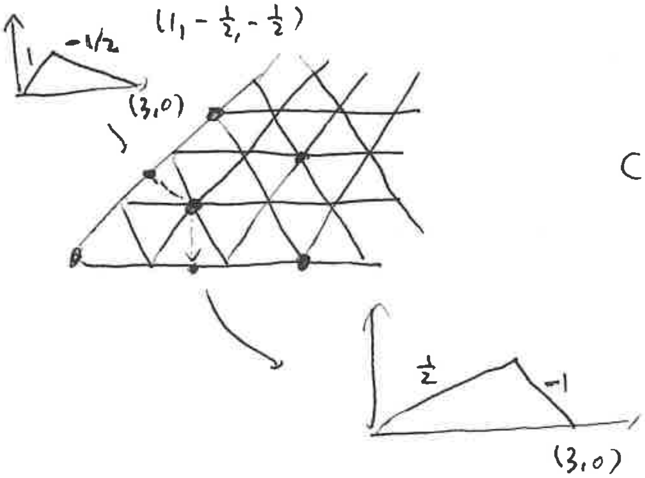
- $\frac{LG}{LG}$ $\quad LG = \bigcup (LG)_\nu$

Newton points $\quad G =$ s.c.

$$\cap$$
$$\mathfrak{a}_G^+ = X_*(T)_{\mathbb{Q}}^+$$

SL_2 :

SL_3 :



$C\ (\mathbb{R}^3)^{\Sigma=0}$

$(\frac{1}{2}, \frac{1}{2}, -1)$

$\mathcal{H} \supset \mathcal{H}_J$ $\qquad J \subset \tilde{I}$ = affine simple refl. of $\tilde{W}$

$\qquad \| \qquad\qquad |W_J| < \infty$

$Sh_{bimon}(N_J \backslash L_J / N_J)$

$LG \supset P_J \twoheadrightarrow L_J \qquad J \leftrightarrow$ simple roots of L_J

eg. $\quad J = \phi \quad , \quad L_J = T \quad , \quad \mathcal{H}_\phi = Sh_{mon}(T)$

SL_2 : $\qquad \tilde{I} = \{s_0, s_1\}$

$$\begin{tikzcd} & \mathcal{H}_{s_0} \arrow[dr] & \\ \mathcal{H}_\phi \arrow[ur] \arrow[dr] & & \mathcal{H} \\ & \mathcal{H}_{s_1} \arrow[ur] & \end{tikzcd}$$

<u>Th</u> (Tao - Travkin) This is a pushout diag in Alg (Cat)

i.e. $\operatorname*{colim}^{\otimes}_{\substack{J \subset \tilde{I} \\ f.t.}} \mathcal{H}_J \xrightarrow{\sim} \mathcal{H}.$

$$hh(A, \overset{\substack{A\text{-bimod} \\ \downarrow}}{M}) = A \otimes_{A \otimes A^{op}} M$$

$$\Rightarrow hh(\mathcal{H}) = hh(\mathcal{H}, \mathcal{H}) \simeq \operatorname*{colim}_J \underbrace{hh(\mathcal{H}_J, \mathcal{H})}_{\text{pages}}$$

Rem. Only need weaker: $\operatorname{colim}_J \mathcal{H} \otimes_{\mathcal{H}_J} \mathcal{H} \xrightarrow{\sim} \mathcal{H}$.

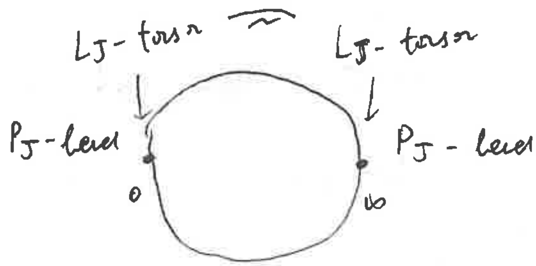
$$hh(\mathcal{H}_J, \mathcal{H}) \simeq Sh_{\mathcal{N}}\left(\boxed{\frac{P_J^+ \backslash LG / P_J^+}{\mathrm{Ad}(L_J)}} \right) =: Y_J$$

$\mathcal{N}$: nilp. s.s. (w.r.t. $L_J \times L_J$-action)

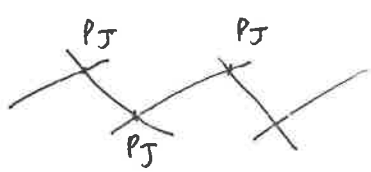
Lusztig "parabolic CS"
horic

Stratification of Y_J

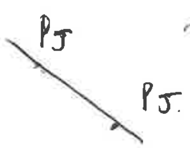
bubbling version of Y_J



Bun_G ("P_J-gluing" )



$\longrightarrow \mathbb{Z}$

$\left| \mathrm{Bun}_G \left(\right. \right.$  $\left. \left. \right) \right| \longleftrightarrow W_J \backslash \widetilde{W} / W_J$

$\longleftrightarrow \left\{ \begin{matrix} (F, F') \\ J\text{-facet} \end{matrix} \right\} / \widetilde{W}$

$J = \phi$, J-facet = alcove in $\mathfrak{a} = X_*(T)_{\mathbb{R}}$

$J = \max$, J-facets are vertices

$|\mathrm{Bun}_G(\infty\text{-chain}, P_J \cdots)| \longleftrightarrow \left\{ (F_n)_{n \in \mathbb{Z}} \right\} / \widetilde{W}$
J-facets

$$\left| \mathrm{Bun}_G \left({}_{P_J}\mathbb{P} \right) \right| \longrightarrow \{ \text{periodic } (F_n) \} / \tilde{W} =: \mathcal{S}_J$$

periodic: $\exists\, w \in \tilde{W}$ s.t. $F_n = w^n F_0, \ \forall n \in \mathbb{Z}$.

$\Rightarrow Y_J$ has a stratification indexed by $\mathcal{S}_J$.

$\mathcal{S}_J$. $J = \phi$:
$$\{ \text{periodic } \underset{\text{alcoves}}{(A_n)} \} / \tilde{W} \longleftrightarrow \tilde{W}$$

may assume $A_0 =$ f.d. alcove.

$$(w^n A_0)_n \longleftarrow\!\shortmid\; w$$

Th (Bédard, Lusztig) canonical bijection $\mathcal{S}_J \simeq [W_J \backslash \tilde{W}]$ ← min length

$$\frac{u}{J} := \underset{\substack{\llcorner \\ \text{standard}}}{(u^n F_J)_n} \longleftarrow\!\shortmid\; u \in [W_J\backslash\tilde{W}]$$

Geom of strata
$$Y_J\left(\frac{u}{J}\right) \xrightarrow{\text{gerbe}} \frac{L_K}{\mathrm{Ad}_u(L_K)}$$

$K \subset J$ (max sub set stable under u)

$$\text{Sheaves}_{\mathcal{N}} \text{ here} \simeq \mathrm{Sh}_{\mathcal{N}}\left(\frac{L_K}{\mathrm{Ad}_u(L_K)} \right) = CS^u(L_K)$$

Vary J. $J \subset J'$.

$$Y_J \longleftarrow \frac{P_{J'}^{+} \backslash LG / P_{J'}^{+}}{\mathrm{Ad}(P_{J'}^{J})} \longrightarrow Y_{J'}$$

$$\begin{array}{ccc} ch_J^{J'} : \; Sh_{\mathcal{N}}(Y_J) & \longrightarrow & Sh_{\mathcal{N}}(Y_{J'}) \\ \wr\wr & & \wr\wr \\ hh(\mathcal{H}_J, \mathcal{H}) & \xrightarrow{\text{can } m} & hh(\mathcal{H}_{J'}, \mathcal{H}) \end{array}$$

Remodeling of apartment

$$\mathfrak{a} \qquad \mathrm{Fac}_J \longleftrightarrow \widetilde{W}/W_J$$

$$\mathcal{B} \qquad \mathrm{Fac}_J(\mathcal{B}) \longleftrightarrow \mathcal{S}_J$$

Def. $\mathcal{B} = \operatorname{colim}_{(E,w)} E$

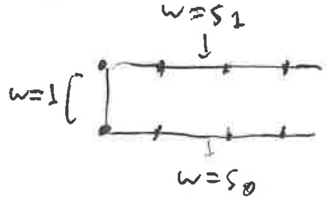
E : relevant aff. subsp. of $\mathfrak{a}$
$\circlearrowleft$
w : restriction of some $\tilde{w} \in \widetilde{W}$.

$(E, w) \xrightarrow{g \in \widetilde{W}} (E', w')$

means $\boxed{gE \subset E'}$ $\quad gwg^{-1}|_{gE} = w'|_{gE}$.

- $\mathcal{B}$ is stratified by J-facets
- $\mathrm{Fac}_J(\mathcal{B}) \simeq \mathcal{S}_J = \{\text{periodic chain } (F_n)\}/\widetilde{W}$
- $\mathcal{B} = \coprod_{\nu \in \mathcal{N}(G)} \mathcal{B}_\nu$

$\underline{SL2}$. $\nu = 0$ $\quad \mathcal{B}_0$



(w has finite order)
$\nu_w = 0$

$\nu = 1$ $\quad \mathcal{B}_1$

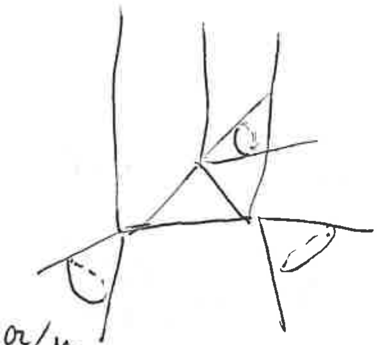
$\simeq \mathfrak{a}/\mathbb{Z}$

$\nu = 2$ $\quad \mathcal{B}_2$ $\quad$ same

$\vdots$ $\quad \mathcal{B}_3$

SL_3

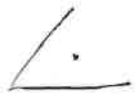
$\nu = 0$ $\quad B_0$



$\mathfrak{a}/\mu_3$

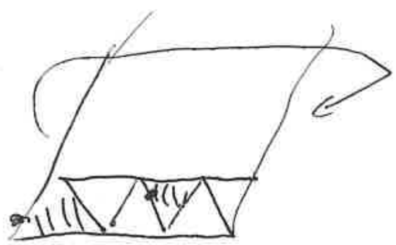
$\times [0, \infty)$

$\nu =$ generic

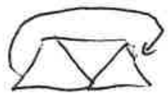


$B_\nu = \mathfrak{a}/X_*(T)$, 2-torus

$\nu = \left(\frac{1}{2}, \frac{1}{2}, -1\right)$, $\quad B_\nu$



essential ~~plact~~ part of B_ν



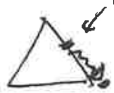
Möbius strip

$B_\nu^\heartsuit \subset B_\nu$

straight element

$\mathrm{Alc}\,(B_\nu^B) \longleftrightarrow \left\{ \begin{array}{l} \text{straight } w \\ \text{with } \nu_w = \nu \end{array} \right\} / \tilde{W}$

$\begin{array}{c} \mathcal{F} \\ \downarrow \\ B \end{array}$: cosheaf of categories



put $Sh_{\mathcal{N}}\left(Y_J\left(\frac{u}{J}\right)\right) \simeq CS^u(L_K)$

Roughly, $\displaystyle\int_B \mathcal{F} \simeq \operatorname*{colim}_J hh(\mathcal{H}_J, \mathcal{H})$

Th. $hh(\mathcal{H})$ has SOD by $\mathcal{N}(G)$ s.t. $hh(\mathcal{H})_\nu \simeq \operatorname*{colim}_{\substack{F \text{ facets} \\ \text{of } B_\nu^B}} CS^u(L_{K(u,J)})$

$\begin{array}{c} \updownarrow \\ \frac{u}{J} \end{array}$ (par. ind of CS)

Ex. $\nu = 0$, $hh(\mathcal{H})_0 \simeq \operatorname*{colim}_{\substack{J \subset \tilde{I} \\ \text{f.t.}}} CS(L_J)$

$\left(\simeq Sh_{\mathcal{N}}\left(\mathrm{Bun}_G(E)^{ss}\right)\right)$

Ex ν = generic

$$hh(\mathcal{H})_\nu \simeq \operatorname*{colim}_{\mathcal{O}/X_*(T)} CS(T) = CS(T)^{X_*(T)}$$

$$\simeq CS\left(\frac{(LT)^\circ}{LT}\right)$$

$$\simeq Sh_{\mathcal{N}}(Bun_G(E)^\nu)$$

$$\wr\!\!\wr$$

$$E \underset{\mathbb{Z}}{\otimes} X_*(T)$$

grad flow by He-Nie

$$B_\nu^\heartsuit \rightleftarrows B_\nu$$

$\subset$ unit(f_{HN}) h.e.

$\boxed{\text{categorical contraction}}$

Lecture 3 (DN) Th (Li-N-Yun) G s.c.

$$hh(\mathcal{H}) \xrightarrow[\sim]{\text{horocycle}} Sh_{\mathcal{N}}(Bun_G(E))$$

genus 1

$hh(\mathcal{H},\mathcal{H}) =$

$Sh_{\mathcal{N}}(I^+\backslash LG/I^+)$

$$\operatorname*{colim}_{J \subset \tilde{I}^+} hh(\mathcal{H}_J, \mathcal{H})$$

$\nearrow\sim$ asymptotics

finite Hecke cat.

$Sh_{\mathcal{N}}(U\backslash L_J/U)$

Reminder $G = SL_2 \subset \bar{G}$ wonderful compactification

$ad - bc = 1$

$G/B \times B^-\backslash G$

$G/N^- \overset{T}{\times} N\backslash G$

punctured normal bundle to boundary strat.

Pass to adj. quotient

$$\frac{G}{G} \xleftarrow{\pi} \frac{G}{B} \xrightarrow{\delta} G/N \overset{T}{\times} N\backslash G \qquad \text{(horocycle)}$$

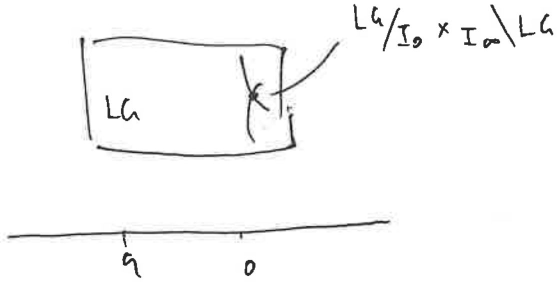
$$\frac{G}{G} \rightsquigarrow G/N^- \overset{T}{\times} N\backslash G \qquad \text{(asymptotics)}$$

$$\boxed{\delta_* \pi^! = \Psi}$$

Solis: loop gp wonderful comp. $LG = G[t, t^{-1}]$, $\overline{LG \rtimes \mathbb{G}_m}$

$$\downarrow$$

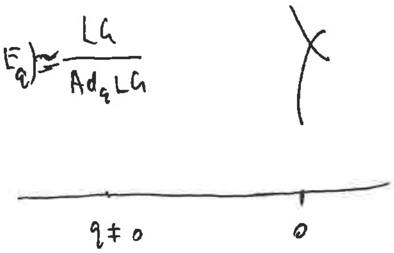
$$\mathbb{A}^1$$



$\big|\ A_{fund}$

<u>Pass to adj. quotient</u> (analytically ... over Tate curve)

$$Bun_G(E_q) \simeq \frac{LG}{Ad_q LG}$$



$\big|\ A_{fund} \hookleftarrow J \qquad \dfrac{P^+_{J,0}\backslash LG / P^+_{J,\infty}}{L_J}$

(global) parahoric char stack

<u>Want to prove</u>:

$$\lim_{\substack{J \subset \tilde{I} \\ \neq}} Sh_N\left(\frac{P^+_{J,0}\backslash LG/P^+_{J,\infty}}{L_J}\right) \xleftarrow[\sim]{\text{induced by } \Psi} Sh_N(Bun_G(E_q))$$

Newton SOD $\longleftrightarrow$ HN SOD

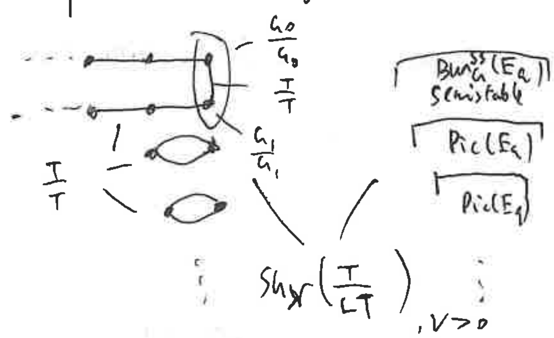
|

~~Mutation~~ (rotation) of Eis/CT SOD

<u>Prop</u>. Ψ commutes with CT. $\Rightarrow$ in fact

<u>Remains</u> to see Ψ induces equiv. on assoc. graded

<u>Cartoon</u> $G = SL_2$



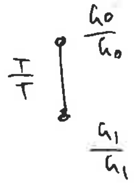
$0 = \nu$

1

2

$\vdots$

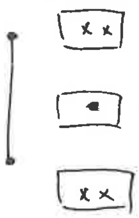
$\nu = 0 \qquad B_\nu^\heartsuit \subset B_\nu$



$\hookleftarrow Bun_G^{ss}(E_q)$

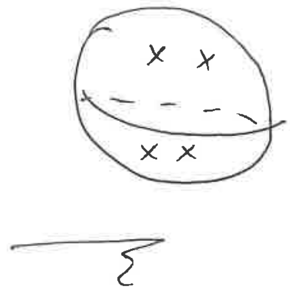
Coarse moduli

$T//W$



T

$T//W$



$\mathbb{P}^1$

Σ

Apps / Ques.

1) $\mathrm{End}(\mathrm{Whit}_{G, E_q})$ ← Li-N-Yun calculated

$\mathcal{N}$-sing. supp. obj. corepresenting Whit. functional

$\mathrm{Hom}(\mathrm{Whit}_{G,E_q}, \mathcal{O}(\check{G}) * \mathrm{Whit}_{G,E_q}) = ?$

2) Th (Fratila - Gunningham - Li)

$Sh_{\mathcal{N}}(Bun_G^{ss}(E_q)) \supset$ cuspidal ell. char. sheaves CS^{ϕ}_{G,E_q}

$$CS^{\phi}_{G,E_q} \simeq CS^{\phi}_{\check{G},E_q}$$

Ex. $G = SL_2$

$\check{G} = PGL_2$.

$*/\mu_2 \times \mu_2$

Prove this via quantum version?

Lecture 4 (ZY) CS on loop Lie algebras (joint with B.-C. Ngô)

Orbit method. (Kirillov, Kostant, ...)

$G_a = Z \subset H$ Heisenberg gp

$$\begin{array}{ccc} \mathcal{O} = \pi^{-1}(\psi) & \subset & \mathfrak{h}^* \\ \text{coadj. orb} \downarrow & & \downarrow \\ \psi & \in & k = (\mathrm{Lie}\, Z)^* \end{array}$$

$H \simeq \mathfrak{h}$

$\chi(WH_\psi) = \psi_Z$ extending by 0

- $\mathbb{1}_{\mathcal{O}} \xrightarrow{FT}$ characters of rep

- $\mathcal{O}$ sympl mfld $\xrightarrow[\text{quantization}]{\text{geom.}}$ $\pi \circlearrowright H = Z \times V$

For p-adic gps, co-adjoint orbits need to be thickened. $\mathbb{O}$

- $\mathbb{1}_{\mathbb{O}} \xrightarrow{FT}$ Lie alg analog of characters

- $\mathbb{O} \xrightarrow[\text{Yu constr.}]{\text{geom quant}} \pi$

- CS sheaf on $\mathbb{O} \subset Lg^* \xrightarrow[\text{def'n}]{FT}$ CS on Lg (supp. on $(Lg)_c$)

$G/F = k((t))$, $k = \bar{k}$, $\mathrm{char}(k) \nmid |W|$.

$g = \mathrm{Lie}\, G$ v.s. $/F$ $\rightsquigarrow LG$, Lg

$g^* = \mathrm{Hom}_F(g, \omega_F)$ Lg^*
$\omega_F \ni f \cdot dt$

Polar sets

Polar data
- Howe factorization
- Yu data
- Levelt - Turritin classif. of formal ∇

Toral case: $T \subset G$ max. torus

$\boxed{(T, \lambda), \quad \lambda \in t^* / (t_c)^\perp \quad \lambda \text{ is } G\text{-regular}}$ Toral polar datum

$T(\mathcal{O}_F) \subset T(F)$
Lie: $t_c \subset t$

$T = T_0 \otimes_k F$ split case

$$\lambda = \left(\frac{\lambda_n}{t^n} + \frac{\lambda_{n-1}}{t^{n-1}} + \cdots + \frac{\lambda_1}{t} + \lambda_0\right)\frac{dt}{t} \qquad \text{where } \lambda_i \in \mathfrak{t}_0^* = (\mathrm{Lie}\, T_0)^*$$

G-regular : $\forall\, \alpha \in \Phi(G,T)$, $\langle d\alpha^\vee, \lambda_i\rangle \neq 0$ for some i.

General polar datum : $M \subset G$ twisted Levi

(M, λ) $\qquad T_M = M/M^{\mathrm{der}}$, $\lambda \in \mathfrak{t}_M^* / \mathfrak{t}_{M,c}^{\perp}$

G-regular

Ex. $(G, 0)$

$PD(G)$ = polar data for G

$\circlearrowleft$

$G(F)$

Polar set. $\qquad c^* = \mathfrak{g}^* /\!/ G$, $\qquad Lc^*$ $\qquad Y_{(M,\lambda)} \subset Lc^*$

(T, λ)

$$\begin{array}{ccc} \lambda + (\mathfrak{t}_c)^{\perp} & \subset & L\mathfrak{g}^* \\ \downarrow & & \downarrow \\ \boxed{Y_{(T,\lambda)}} & \subset & Lc^* \end{array}$$

Fact. This is a closed subscheme of f.p.

(M, λ)

$$\begin{array}{c} \lambda + (L\mathfrak{m}^*)_{\mathrm{t.n.}} \\ \downarrow \\ Y_{(M,\lambda)} \subset Lc^* \end{array}$$

Ex. SL_2 $\qquad LC^* = k((t))(dt)^2$

$T = \mathbb{G}_m$ $\qquad \lambda = \left(\frac{\lambda_n}{t^n} + \cdots + \lambda_0\right)\frac{dt}{t}$, $\qquad$ depth $n \geq 0$

$$Y_{(T,\lambda)} = \left\{ \left(\frac{\alpha_{2n}}{t^{2n}} + \cdots + \frac{\alpha_n}{t^n} + \mathcal{O}_F \cdot t^{-n+1} \right) \left(\frac{dt}{t}\right)^2 \right\}$$

α_i determined by λ

$T = (R_{E/F}\mathbb{G}_m)^{N_m = 1}$ $\qquad$ depth $n - \frac{1}{2}$ $\quad (n \geq 1)$

$$Y_{(T,\lambda)} = \left\{ \left(\frac{\alpha_{2n-1}}{t^{2n-1}} + \cdots + \frac{\alpha_n}{t^n} + \mathcal{O}_F\, t^{-n+1} \right) \left(\frac{dt}{t}\right)^2 \right\}$$

$$Y_{(G,0)} = \left\{ t\, \mathcal{O}_F \left(\frac{dt}{t}\right)^2 \right\}$$

Prop $\{Y_{(M,\lambda)}\}_{(M,\lambda) \in PD(G)/G(F)}$ give a partition of LC^*

thickened coadj. orbit

$$\begin{array}{ccc} Z_{(M,\lambda)} & \subset & Lg^* \\ \downarrow & \lrcorner & \downarrow \\ Y_{(M,\lambda)} & \subset & LC^* \end{array}$$

Polar data v.s. Yu data

(M, λ) $\qquad M = G^0$

$$\begin{array}{c} \cap \\ C_G\left(\text{truncations of } \lambda\right) = G^1 \\ \cap \\ \vdots \end{array}$$

C.S. on Lie algebra (Lusztig)

$G / k = \bar{k}$, $\qquad OS(g^*) \xleftrightarrow{FT} CS(g)$

orbital sheaves

Coadj. orb. $\mathcal{O} \subset \mathfrak{g}^*$, $\mathcal{L}$: G-eq. simple local system on $\mathcal{O}$

$\rightsquigarrow IC(\bar{\mathcal{O}}, \mathcal{L}) \xrightarrow{FT} \mathcal{F}_{(\mathcal{O},\mathcal{L})} \in CS(\mathfrak{g})$

C.S. on $L\mathfrak{g}$:

$OS(L\mathfrak{g}^*)$

(T, λ) polar datum, assume T = elliptic. $T(F)^\circ = T(\mathcal{O}_F)$

$\widetilde{\mathrm{Ch}}_{\mathcal{L}}\, Z_{(T,\lambda)} \in OS(L\mathfrak{g}^*)$

$$\begin{array}{ccc} [Z_{(T,\lambda)}/LG] & \simeq & [(\lambda + \mathfrak{t}_c^{\perp})/LT] \\ | & & | \qquad \backslash \\ M & & (L\mathfrak{m}^*)_{tn} \quad LM \end{array}$$

General $(M, \lambda) \rightsquigarrow (M, \lambda, \underbrace{L, \mathcal{O}, \mathcal{L}}) = \xi$

cuspidal triple for LM

L : Levi of parahoric $Q \subset LM$

$\mathcal{O} \subset \mathfrak{l}$ nilp orb

$\mathcal{L}$ cusp. local sys. on $\mathcal{O}$.

$$\begin{tikzcd} {[\widetilde{Z}_{(M,\lambda)}/LG]} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \lambda + \overset{\mathcal{O}+}{\mathfrak{q}^{\perp}}/Q \arrow[r] \arrow[d] & {[\mathcal{O}/L]} \\ Z_{(M,\lambda)}/LG \simeq & \lambda + (L\mathfrak{m}^*)_{tn}/LM & \end{tikzcd}$$

$\mathcal{L}$ over $[\mathcal{O}/L]$

$\mathcal{L} \rightsquigarrow \rho_\xi$

$OS(L\mathfrak{g}^*) = \langle \rho_\xi : \text{all } \xi \rangle \subset D_{LG}(L\mathfrak{g}^*)$

$(G, 0, I, T, \{0\}, \text{triv}) = \xi$

$\mathcal{S}_\xi$ on $(Lg^*)_{tn}/LG$ is the affine Springer sheaf of BKV

$CS(Lg) := FT(OS(Lg^*))$.

<u>Th</u> (Bernstein decomp)

$$CS(Lg) = \bigoplus_{\eta = (M,\lambda,L,\mathcal{O},\mathcal{L})/G(F)} CS_\eta(Lg)$$

<u>Fact</u> $\widetilde{Z}_{(M,\lambda)}$ is symplectic $\boxed{T \text{ elliptic } (T,\lambda), \ Z_{(T,\lambda)} \text{ symplectic.}}$

$(M,\lambda) \rightsquigarrow$ Yu datum $\rightsquigarrow J$ (choose lag...)

$$\widetilde{Z}_{(T,\lambda)} = T^*_\lambda(LG/J)$$

Yu's rep ~ functions on $LG/(J,\lambda)$

<u>Global Hitchin spaces</u>

X, x

$$\mathcal{M}_{(M,\lambda,\alpha)} = \left(T^* \mathrm{Bun}_{G,\infty x} \times \widetilde{Z}_{(M,\lambda)}\right) /\!\!/\!\!/ L_x G$$

α parahoric in LM

<u>expectation</u>: Hitchin map is a comp. integrable sys.